ILIYA VALIZADEH

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SUMMARY

Third-year data science student at York University who builds data pipelines, predictive models and NLP tools in Python and SQL, then hands them to the people who use them. Open to a Winter 2027 co-op or a Summer 2027 internship.

EDUCATION

York University, BSc Data Science GPA 8.83/9.0 (A+) | Toronto, ON | Expected April 2028

Relevant coursework: Regression Analysis, Numerical Methods, Simulation and Monte Carlo Methods, Principles and Techniques of Data Science, Data Structures for Data Science, Data and Information Management (some Fall 2026 in progress).

Awards and certifications: Dean's Honour Roll (2026-27); Databricks Certified Data Engineer Associate (2026)

TECHNICAL SKILLS

Data engineering: Python (pandas, NumPy), SQL, Databricks (Spark SQL, Delta Lake), DuckDB, PostgreSQL, Parquet, star schema design, idempotent loads and backfills, data transformation, cleaning, validation and quarantine

Machine learning and AI: scikit-learn, LightGBM, model calibration and evaluation, retrieval-augmented generation (FAISS, BM25, sentence-transformers), LLM API (Gemini), NLP

Tools: Git, GitHub Actions CI, pytest, Tableau, Power BI, Excel, Pyodide, GitHub Pages

EXPERIENCE

Undergraduate Data Science Researcher (Work/Study Program), York University Toronto | May to Aug 2026

- Built a Python (pandas) pipeline that merges about 20 raw Excel/CSV files (60,000+ rows of Statistics Canada, municipal and operational data) into one analysis-ready table with standard headers and dates and no duplicate keys.
- Added checks for duplicate keys, missing join keys and out-of-range percentages that send failing rows to a review file.
- Cut a weekly 4 to 5 hour manual Excel refresh to under 5 minutes. Demoed each update of the live Tableau dashboard to my supervisor and a 6 to 8 person research team, then trained them to run it on their own; still in use after handoff.

Clinical Data Research Assistant (Volunteer), Centre for Addiction and Mental Health (CAMH) Toronto | Jan 2025 to present

- Link and validate longitudinal data for a youth and young adult mood and anxiety cohort (800+ participants, up to 5 visits, 4,500+ records, 150+ variables) that the PI, lab statistician and graduate students use for outcome analysis and grant reporting.
- Wrote a reusable Python (pandas) script that repairs 25 to 40 broken participant-ID links per batch, fixes misordered visit dates and resolves duplicates. Review the flagged rows monthly with the study coordinator and statistician.

Co-President, York University Data Science Community Toronto | Sept 2026 to present

- Co-lead an 8-person executive team for a 250+ member club; co-present hands-on Python, SQL and machine learning workshops to 35+ students each and run flagship events for 70+.

Server, Mandarin Restaurant Toronto | Sept 2024 to present

PROJECTS

Credit-Risk Scorecard, LightGBM default model (0.769 ROC-AUC) github.com/Iliya-Valizadeh/credit-risk-scorecard

Python, pandas, scikit-learn, LightGBM, SHAP, PostgreSQL, pytest, GitHub Actions

- Turned scores into an approval policy: declining the riskiest 20% caught 53.2% of applicants who later had payment difficulties, and expected loss under an assumed 45% LGD came within 2% of the observed figure.
- Found class weighting had pushed the average predicted default rate to 40% against an actual 8%, and brought it back to 8% with isotonic recalibration (Brier 0.185 to 0.067).

NYC Rideshare Data Pipeline github.com/Iliya-Valizadeh/nyc-rideshare-congestion

Python, DuckDB, SQL, Parquet, pytest

- Validated 532.2 million trip records from 27 months, quarantining 114,113 bad rows (0.02%) with a reason code each, and aggregated the rest into a star schema with a 242.4 million row fact table: one row per hour, platform and zone pair.

Bank Filings Analyst, LLM question answering github.com/Iliya-Valizadeh/bank-filings-rag

Python, Hugging Face sentence-transformers, FAISS, BM25, PyMuPDF, LLM API (Gemini), pytest, GitHub Actions

- Answers questions over RBC's 250-page annual report and cites the source page for each claim. Built as a Python package with separate modules for ingestion, chunking, indexing, retrieval and answer generation.
- Compared 3 chunking and 3 search methods on a hand-checked answer key: whole pages with hybrid search found the right page for 7 of 12 questions, against 3 of 12 for fixed 180-word chunks.

Second Look, bank statement checker iliya-valizadeh.github.io/second-look

Python, Pyodide, GitHub Pages

- Public web app that runs Python in the browser (Pyodide) and flags recurring charges in bank statements; on synthetic statements it found 95.9% of planted recurring charges at 86.1% precision, against 77.7% and 24.1% for a baseline.